Mugilan Mariappan

PhD in Computing Science
Systems & Parallel Computing | SFU

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Research Interests

Streaming graph processing, graph databases, and GNN infrastructure, with emphasis on parallelism, incremental computation, and dynamic data structures.

Education

Ph.D. in Computing Science — Burnaby, Canada
Simon Fraser University — 2018–2026
Dissertation: Dependency-Driven Incremental Processing of Streaming Graphs
Advisor: Dr. Keval Vora | **Defended:** October 2026

B.E. in Computer Science — Coimbatore, India
Coimbatore Institute of Technology (GPA: 8.5/10) — 2010–2014

Publications

1. GraphBolt: Dependency-Driven Synchronous Processing of Streaming Graphs — EuroSys 2019
   Mugilan Mariappan and Keval Vora.
   European Conference on Computer Systems (EuroSys), 2019. | 183 citations
2. DZiG: Sparsity-Aware Incremental Processing of Streaming Graphs — EuroSys 2021
   Mugilan Mariappan, Joanna Che, and Keval Vora.
   European Conference on Computer Systems (EuroSys), 2021. | 79 citations
3. Anti-Vertex for Neighborhood Constraints in Subgraph Queries — GRADES-NDA 2022
   Kasra Jamshidi, Mugilan Mariappan, and Keval Vora.
   ACM SIGMOD Workshop on GRADES-NDA, 2022. | 7 citations
4. Threshold Strategy for Leaking Corner-Free Hamilton-Jacobi Reachability — CDC 2025
   Chong He, Mugilan Mariappan, Keval Vora, and Mo Chen.
   IEEE Conference on Decision and Control (CDC), 2025. | 1 citation

Research & Industry Experience

Simon Fraser University — Burnaby, Canada
Graduate Research Assistant, PDC Lab (PDCL) — 2018–2026

- Built **GraphBolt**, a dependency-driven streaming graph engine that makes incremental graph analytics with BSP guarantees practical on constantly evolving graphs, without exhaustive state tracking. Achieves a median **18× speedup** (up to 232×) over recomputation from scratch for **billion-edge graphs on a single machine**.
- Built **DZiG**, a sparsity-aware engine that pushes the **effectiveness** of dependency-driven incremental processing to **update batches orders of magnitude larger** than prior systems. Designed an adaptive executor that uses a linear-regression cost model to switch strategies at runtime, across algorithms, graphs, and update patterns.
- Built **Olfin**, a concurrent graph ingestion system for multi-source update streams with tunable ordering guarantees, using lockless, linearizable graph update operations. Speeds up GraphBolt's ingestion by **8.5–43×**.
- Formalized matching semantics and prototyped grammar extensions to Cypher for expressing negative neighborhood constraints in subgraph queries, enabling automatic reasoning and efficient query evaluation.
- Optimized feature-aware CUDA kernels for GCN, GAT, and GGNN training, improving memory coalescing.

LinkedIn Graph Systems Lab — Sunnyvale, USA
Graduate Research Intern — Sep 2022–Dec 2022

- Benchmarked the proprietary **Liquid Graph Database** against the complex-query workloads of the LDBC Social Network Benchmark, diagnosing root causes of slow execution and identifying optimizations.
- Surfaced design constraints for graph query processing with mentors at LinkedIn and **Microsoft Research (Gray Systems Lab)**, informing a rearchitecture of Liquid's query processing over the next two years.

athenaHealth — Chennai, India
Software Engineer — May 2014–Jul 2017

- Rearchitected the secure messaging module of the athenaText iOS app, reducing crashes by **50%** and eliminating UI stalls during message sync and refresh.
- Maintained and extended the indexing and ranking components of search for Epocrates, a clinical reference app used by over 50% of US physicians.

Research Software

GraphBolt/DZiG — Open source | 150 stars, 25 forks on GitHub

- End-to-end C++ system for streaming graph analytics: a stream ingestor, a dynamic graph store, and general-purpose incremental graph engines.
- Programming model and runtime for writing new iterative graph algorithms that run incrementally and in parallel with BSP guarantees, without exhaustive edge-level state tracking.

Technical Skills

Expertise: Parallel systems (design, debugging, optimization), incremental processing, dynamic data structures
Technologies: C++, CUDA, OpenMP, pthreads, Cilk Plus, perf, Python, Java, SQL, Cypher

Talks

GraphBolt: Dependency-Driven Synchronous Processing of Streaming Graphs, EuroSys, Dresden — 2019
Incremental Graph Processing With GraphBolt and DZiG, Microsoft Gray Systems Lab, Sunnyvale — 2022
Incremental Graph Processing With GraphBolt and DZiG, LinkedIn, Mountain View — 2022

Teaching & Mentoring

Teaching Assistant (SFU): Distributed Systems (CMPT 431), Operating Systems (CMPT 300).
Mentoring (PDCL): 6 students in systems research and performance engineering.

Honors & Awards

Helmut & Hugo Eppich Family Graduate Scholarship, SFU — 2022
Graduate Fellowship, SFU — 2018, '20, '22, '23
CMPT Graduate Fellowship, SFU — 2019, '21, '22
Best Poster Award (with Kasra Jamshidi), SFU CS Research Day — 2022
Best Poster Award (with Joanna Che), SFU CS Research Day — 2019
Special Graduate Entrance Scholarship, SFU — 2018

Professional Service

- **Reviewer:** IEEE Transactions on Knowledge and Data Engineering (TKDE), 2026.
- **External Reviewer:** EuroSys (2020, '24, '27); USENIX ATC (2020, '24); SOSP (2023); SIGMOD (2023); ICDCS (2023); PACT (2020); IEEE TPDS (2023); IEEE TKDE (2022).
- **Departmental (SFU):** CS Councillor, Graduate Student Society (2018–19, 2022); Secretary, CS Graduate Student Association (2018–19).